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United States Patent	12400784
Kind Code	B2
Date of Patent	August 26, 2025
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Inductor device

Abstract

An inductor device includes a first trace, a second trace, and a connection member. The first inductor includes a first trace and a second trace. The shape of one of the first trace and the second trace is spiral. The second inductor includes a third trace and a fourth trace. The connection member is configured to couple the first inductor and the second inductor.

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Appl. No.:	17/647466
Filed:	January 09, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230055317 A1	Feb. 23, 2023

Foreign Application Priority Data

TW	110130908	Aug. 20, 2021
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Publication Classification

Int. Cl.: H01F27/28 (20060101); H01F17/00 (20060101); H01F27/29 (20060101)

U.S. Cl.:

CPC H01F27/29 (20130101); H01F17/0013 (20130101); H01F2017/0073 (20130101)

Field of Classification Search

CPC: H01F (27/29)

USPC: 336/200; 336/232; 336/225

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Background/Summary

RELATED APPLICATIONS

(1) This application claims priority to and the benefit of Taiwan Application Serial Number 110130908, filed on Aug. 20, 2021, the entire contents of which are incorporated herein by reference as if fully set forth below in its entirety and for all applicable purposes.

BACKGROUND

Field of Invention

(2) The present disclosure relates to an electronic device. More particularly, the present disclosure relates to an inductor device.

Description of Related Art

(3) The various types of inductors according to the prior art have their advantages and disadvantages. For example, a spiral inductor has a higher Q value and a larger mutual inductance. However, coupling is occurred between a spiral inductor and other devices. For an eight-shaped inductor/transformer which has two sets of coils, the coupling between the two sets of coils is relatively low. However, an eight-shaped inductor/transformer occupies a larger area in a device. In addition, when an eight-shaped inductor/transformer is designed to be a symmetrical structure, paristic capacitances of differential signals of the eight-shaped inductor/transformer increase substantially due to traditional crossing structures. Therefore, the scopes of application of the above inductor/transformer are limited.

SUMMARY

- (4) The foregoing presents a simplified summary of the disclosure in order to provide a basic understanding to the reader. This summary is not an extensive overview of the disclosure and it does not identify key/critical elements of the present disclosure or delineate the scope of the present disclosure. Its sole purpose is to present some concepts disclosed herein in a simplified form as a prelude to the more detailed description that is presented later.
- (5) One aspect of the present disclosure is to provide an inductor device. The inductor device comprises a first inductor, a second inductor, and a connection portion. The first inductor comprises a first trace and a second trace. The second inductor comprises a third trace and a fourth trace. A shape of one of the first trace and the second trace comprises a spiral. The third trace and the fourth trace are symmetrical to each other. The connection portion is configured to couple to the first inductor and the second inductor.
- (6) Therefore, based on the technical content of the present disclosure, the parasitic capacitance can be decrease owing to the disposition of the inductor device of the present disclosure. The differential signals of the symmetrical circuit are inputted at the same side, for example, the positive voltage signal or the negative voltage signal (P or N) are inputted at the same side. A new design, which adopts the positive voltage signals which are adjacent to each other (P and P are adjacent to each other), can avoid the capacitance generated by the positive voltage signal and the negative voltage signal which are adjacent to each other (N/P are adjacent to each other) in a traditional manner. Therefore, the parasitic capacitance decreases owing to the disposition of the inductor device of the present disclosure, such that the quality factor (Q factor) enhances and the summarization of the structure increases.
- (7) It is to be understood that both the foregoing general description and the following detailed description are by examples, and are intended to provide further explanation of the invention as claimed.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings are included to provide a further understanding of the invention, and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention. In the drawings,
- (2) FIG. 1 depicts a schematic diagram of an inductor device according to one embodiment of the present disclosure;
- (3) FIG. 2 depicts a schematic diagram of a portion structure of the inductor device shown in FIG. 1 according to one embodiment of the present disclosure;
- (4) FIG. 3 depicts a schematic diagram of a portion structure of the inductor device shown in FIG. 1 according to one embodiment of the present disclosure;
- (5) FIG. 4 depicts a schematic diagram of a portion structure of the inductor device shown in FIG. 1 according to one embodiment of the present disclosure;
- (6) FIG. 5 depicts a schematic diagram of an inductor device according to one embodiment of the present disclosure;
- (7) FIG. 6 depicts a schematic diagram of an inductor device according to one embodiment of the present disclosure;
- (8) FIG. 7 depicts a schematic diagram of an inductor device according to one embodiment of the present disclosure; and
- (9) FIG. 8 depicts a schematic diagram of experimental data of an inductor device according to one embodiment of the present disclosure.
- (10) According to the usual mode of operation, various features and elements in the figures have

not been drawn to scale, which are drawn to the best way to present specific features and elements related to the disclosure. In addition, among the different figures, the same or similar element symbols refer to similar elements/components.

DESCRIPTION OF THE EMBODIMENTS

(11) To make the contents of the present disclosure more thorough and complete, the following illustrative description is given with regard to the implementation aspects and embodiments of the present disclosure, which is not intended to limit the scope of the present disclosure. The features of the embodiments and the steps of the method and their sequences that constitute and implement the embodiments are described. However, other embodiments may be used to achieve the same or equivalent functions and step sequences.

(12) Unless otherwise defined herein, scientific and technical terminologies employed in the present disclosure shall have the meanings that are commonly understood and used by one of ordinary skill in the art. Unless otherwise required by context, it will be understood that singular terms shall include plural forms of the same and plural terms shall include the singular.

Specifically, as used herein and in the claims, the singular forms “a” and “an” include the plural reference unless the context clearly indicates otherwise.

(13) FIG. 1 depicts a schematic diagram of an inductor device **1000** according to one embodiment of the present disclosure. As shown in the figure, the inductor device **1000** includes a first inductor **1100**, a second inductor **1200**, and a connection portion **1300**. The first inductor **1100** includes a first trace **1110** and a second trace **1120**. The second inductor **1200** includes a third trace **1210** and a fourth trace **1220**. As shown in the figure, the connection portion **1300**, which is located on a central area, can be configured to couple to the first inductor **1100** and the second inductor **1200**.

(14) For facilitating the understanding of the inductor device **1000** in FIG. 1, reference is now made to both FIG. 2 to FIG. 4. FIG. 2 to FIG. 4 depict schematic diagrams of a portion structure of the inductor device **1000** shown in FIG. 1 according to one embodiment of the present disclosure. As shown in FIG. 2, in one embodiment, the shape of one of the portion trace **1112** of the first trace **1110** and the portion trace **1121** of the second trace **1120** can be spiral. For example, the shape of one of the portion trace **1112** and the portion trace **1121** can be spiral, or the shapes of both of the portion trace **1112** and the portion trace **1121** are spiral.

(15) As shown in FIG. 3, in one embodiment, the third trace **1210** and the fourth trace **1220** are symmetrical to each other. Specifically, the third trace **1210** and the fourth trace **1220** are symmetrical to each other on the basis of the central area of the inductor device **1000**.

(16) In one embodiment, the portion structure of the inductor device **1000** shown in FIG. 2 is located on a first layer, the portion structure of the inductor device **1000** shown in FIG. 3 is located on a second layer, and the portion structure of the inductor device **1000** shown in FIG. 4 is located on a third layer. For example, referring to FIG. 2, the portion trace **1112** of the first trace **1110**, the portion trace **1121** of the second trace **1120**, and the portion connection portion **1320** of the connection portion **1300** are located on the first layer.

(17) In addition, referring to FIG. 3, the portion trace **1111** of the first trace **1110**, the portion trace **1122** of the second trace **1120**, the third trace **1210**, and the fourth trace **1220** are located on the second layer. Besides, referring to FIG. 4, the first sub-connection portion **1310** of the connection portion **1300** is located on the third layer. Furthermore, the third layer shown in FIG. 4 further includes a plurality of connection members **1140**, **1160**, **1240**, **1260**, **1270**, **1290**, input/output terminals **1230**, **1250**, and a center-tapped terminal **1280**. The above-mentioned structures will be described in detail as shown below. As described above, all structures of the inductor device **1000** of the present disclosure are located on three layers. Compared with the traditional inductor device adopting four layers for including all structures, the inductor device **1000** of the present disclosure can save area, volume, and so on, and the inductor device **1000** of the present disclosure merely needs three layers for including all structures, such that the design of the circuit becomes easier.

(18) In one embodiment, the first trace **1110** can be the first spiral trace **1110**, and the first spiral

trace **1110** is disposed on the first layer and the second layer. For example, the portion trace **1112** of the first spiral trace **1110** is disposed on the first layer shown in FIG. 2, and the portion trace **1111** of the first spiral trace **1110** is disposed on the second layer shown in FIG. 3.

(19) In another embodiment, the first inductor **1100** further includes a first input/output terminal **1130** and a first connection member **1140**. The first input/output terminal **1130** is disposed on the first layer shown in FIG. 2, and the first connection member **1140** is disposed on the third layer shown in FIG. 4. With respect to the connection structure, the node A which is disposed on the first input/output terminal **1130** shown in FIG. 2 is coupled to the node A which is disposed on the first connection member **1140** shown in FIG. 4. Subsequently, the node B of the first connection member **1140** is coupled to the node B which is disposed on the portion trace **1111** of the first spiral trace **1110** shown in FIG. 3. After that, the node C of the portion trace **1111** of the first spiral trace **1110** is coupled to the node C which is disposed on the portion trace **1112** of the first spiral trace **1110** shown in FIG. 2.

(20) In one embodiment, the second trace **1120** can be the second spiral trace **1120**, and the second spiral trace **1120** is disposed on the first layer and the second layer. For example, the portion trace **1121** of the second spiral trace **1120** is disposed on the first layer shown in FIG. 2, and the portion trace **1122** of the second spiral trace **1120** is disposed on the second layer shown in FIG. 3.

(21) In another embodiment, the first inductor **1100** further includes a second input/output terminal **1150** and a second connection member **1160**. The second input/output terminal **1150** is disposed on the first layer shown in FIG. 2, and the second connection member **1160** is disposed on the third layer shown in FIG. 4. With respect to the connection structure, the second input/output terminal **1150** shown in FIG. 2 is coupled to the portion trace **1121** of the second spiral trace **1120** which is also shown in FIG. 2. The node D of the portion trace **1121** of the second spiral trace **1120** is coupled to the node D which is disposed on the portion trace **1122** of the second spiral trace **1120** shown in FIG. 3. Subsequently, the node E of the portion trace **1122** of the second spiral trace **1120** is coupled to the node E which is disposed on the second connection member **1160** shown in FIG. 4.

(22) Next, the node F of the second connection member **1160** is coupled to the node F which is disposed on the portion trace **1121** of the second spiral trace **1120** shown in FIG. 2. The node G of the portion trace **1121** of the second spiral trace **1120** is coupled to the node G of the connection member **1330** shown in FIG. 3. Subsequently, the node H of the connection member **1330** is coupled to the node H which is disposed on the first sub-connection portion **1310** of the connection portion **1300** shown in FIG. 4. The node I of the first sub-connection portion **1310** of the connection portion **1300** is coupled to the node I which is disposed on the portion trace **1112** of the first spiral trace **1110** shown in FIG. 2.

(23) In one embodiment, the second inductor **1200** further includes a third input/output terminal **1230** and a third connection member **1240**. The third input/output terminal **1230** is disposed on the third layer shown in FIG. 4, and the third connection member **1240** is disposed on the third layer shown in FIG. 4. With respect to the connection structure, the node J which is disposed on the third input/output terminal **1230** shown in FIG. 4 is coupled to the node J which is disposed on the fourth trace **1220** shown in FIG. 3. The fourth trace **1220**, which is located on the second layer, is coupled through the third connection member **1240** to the fourth trace **1220**, which is also located on the second layer, in an interlaced manner.

(24) In another embodiment, referring to FIG. 1, the portion trace **1121** of the second spiral trace **1120** located on the first layer and the fourth trace **1220** located on the second layer are overlapped to each other partially.

(25) In one embodiment, the second inductor **1200** further includes a fourth input/output terminal **1250** and a fourth connection member **1260**. The fourth input/output terminal **1250** is disposed on the third layer shown in FIG. 4, and the fourth connection member **1260** is disposed on the third layer shown in FIG. 4. With respect to the connection structure, the node K which is disposed on

the fourth input/output terminal **1250** shown in FIG. **4** is coupled to the node K which is disposed on the fourth trace **1220** shown in FIG. **3**. The fourth trace **1220** is coupled through the fourth connection member **1260** to the fourth trace **1220** which is also located on the second layer in an interlaced manner. In another embodiment, the third connection member **1240** and the fourth connection member **1260** are located at different sides of the second inductor **1200**. For example, the third connection member **1240** and the fourth connection member **1260** are located at the lower side and the upper side of the second inductor **1200** respectively.

(26) In another embodiment, in the central area (e.g. the center of the structure shown in the figure) of the inductor device **1000**, the third trace **1210** which is disposed on the second layer shown in FIG. **3** is coupled to the fourth trace **1220** which is also located on the second layer.

(27) In one embodiment, the second inductor **1200** further includes a fifth connection member **1270**, and the fifth connection member **1270** is disposed on the third layer shown in FIG. **4**. With respect to the connection structure, the third trace **1210** shown in FIG. **3** is coupled through the fifth connection member **1270** shown in FIG. **4** to the third trace **1210** which is also located on the second layer in an interlaced manner.

(28) In another embodiment, referring to FIG. **1**, the portion trace **1112** of the first spiral trace **1110** located on the first layer and the third trace **1210** located on the second layer are overlapped to each other partially.

(29) In one embodiment, the second inductor **1200** further includes a center-tapped terminal **1280**, and the center-tapped terminal **1280** is disposed on the third layer shown in FIG. **4**. With respect to the connection structure, the node L which is disposed on the center-tapped terminal **1280** shown in FIG. **4** is coupled to the node L which is disposed on the third trace **1210** shown in FIG. **3**.

(30) In another embodiment, referring to FIG. **1**, the center-tapped terminal **1280** is symmetrical to the third input/output terminal **1230** and the fourth input/output terminal **1250** on the basis of the central area (e.g. the center of the structure shown in the figure) of the inductor device **1000**.

(31) In one embodiment, the second inductor **1200** further includes a sixth connection member **1290**, and the sixth connection member **1290** is disposed on the third layer shown in FIG. **4**. With respect to the connection structure, the third trace **1210** in FIG. **3** is coupled through the sixth connection member **1290** shown in FIG. **4** to the third trace **1210** which is also located on the second layer in an interlaced manner. In another embodiment, the fifth connection member **1270** and the sixth connection member **1290** are disposed at different sides of the second inductor **1200**. For example, the fifth connection member **1270** and the sixth connection member **1290** are located at the upper side and the lower side of the second inductor **1200** respectively.

(32) In another embodiment, the connection portion **1300** further includes a second sub-connection portion **1320**, and the second sub-connection portion **1320** is disposed on the first layer shown in FIG. **2**. As shown in FIG. **1**, in the central area (e.g. the center of the structure shown in the figure) of the inductor device **1000**, the third trace **1210** located on the second layer is coupled through the second sub-connection portion **1320** to the fourth trace **1220** located on the second layer. It is noted that the present disclosure is not limited to the structure as shown in FIGS. **1** to **4**, and it is merely an example for illustrating one of the implements of the present disclosure.

(33) FIG. **5** depicts a schematic diagram of an inductor device **1000A** according to one embodiment of the present disclosure. Compared with the inductor device **1000** shown in FIG. **1**, the structure disposition of the inductor device **1000A** shown in FIG. **5** is different, which will be described in detail as below.

(34) Referring to FIG. **5**, the structure between the first trace **1110A** and the third trace **1210A** of the inductor device **1000A** is adjusted, so as to make the fifth connection member **1270A** located at the upper-left corner in the figure to be disposed on the first layer, and make the sixth connection member **1290A** located at the lower-left corner in the figure to be disposed on the first layer.

(35) Besides, the structure between the second trace **1120A** and the fourth trace **1220A** of the inductor device **1000A** is adjusted, so as to make the fourth connection member **1260A** located at

the upper-right corner in the figure to be disposed on the first layer, and make the third connection member **1240A** located at the lower-right corner in the figure to be disposed on the first layer. It is noted that the element in FIG. 5, whose symbol is similar to the symbol of the element in FIG. 1, has similar structure feature in connection with the element in FIG. 1. Therefore, a detail description regarding the structure feature of the element in FIG. 5 is omitted herein for the sake of brevity. In addition, the present disclosure is not limited to the structure as shown in FIG. 5, and it is merely an example for illustrating one of the implements of the present disclosure.

(36) FIG. 6 depicts a schematic diagram of an inductor device **1000B** according to one embodiment of the present disclosure. Compared with the inductor device **1000** shown in FIG. 1, the structure disposition of the inductor device **1000B** shown in FIG. 6 is different, which will be described in detail as below.

(37) Referring to FIG. 6, the structure between the first trace **1110B** and the third trace **1210B** of the inductor device **1000B** is adjusted, so as to make the fifth connection member **1270B** located at the upper-left corner in the figure to be disposed on the first layer, and make the sixth connection member **1290B** located at the lower-left corner in the figure to be disposed on the first layer.

(38) Besides, the structure between the second trace **1120B** and the fourth trace **1220B** of the inductor device **1000B** is adjusted, so as to make the fourth connection member **1260B** located at the upper-right corner in the figure to be disposed on the first layer, and make the third connection member **1240B** located at the lower-right corner in the figure to be disposed on the first layer. It is noted that the element in FIG. 6, whose symbol is similar to the symbol of the element in FIG. 1, has similar structure feature in connection with the element in FIG. 1. Therefore, a detail description regarding the structure feature of the element in FIG. 6 is omitted herein for the sake of brevity. In addition, the present disclosure is not limited to the structure as shown in FIG. 6, and it is merely an example for illustrating one of the implements of the present disclosure.

(39) FIG. 7 depicts a schematic diagram of an inductor device **1000C** according to one embodiment of the present disclosure. Compared with the inductor device **1000** shown in FIG. 1, the structure disposition of the inductor device **1000C** shown in FIG. 7 is different, which will be described in detail as below.

(40) Referring to FIG. 7, the structure between the first trace **1110C** and the third trace **1210C** of the inductor device **1000C** is adjusted, so as to make the fifth connection member **1270C** located at the upper-left corner in the figure to be disposed on the first layer, and make the sixth connection member **1290C** located at the lower-left corner in the figure to be disposed on the first layer.

(41) Besides, the structure between the second trace **1120C** and the fourth trace **1220C** of the inductor device **1000C** is adjusted, so as to make the fourth connection member **1260C** located at the upper-right corner in the figure to be disposed on the first layer, and make the third connection member **1240C** located at the lower-right corner in the figure to be disposed on the first layer.

(42) In addition, the input/output terminals **1230C**, **1250C** of the inductor device **1000C** can be disposed on the first layer, and the input/output terminals **1230C**, **1250C** are coupled to the third sub-trace **1210C**. Furthermore, the center-tapped terminal **1280C** of the inductor device **1000C** can be disposed on the first layer, and the center-tapped terminal **1280C** is coupled to the fourth sub-trace **1220C**. It is noted that the element in FIG. 7, whose symbol is similar to the symbol of the element in FIG. 1, has similar structure feature in connection with the element in FIG. 1. Therefore, a detail description regarding the structure feature of the element in FIG. 7 is omitted herein for the sake of brevity. In addition, the present disclosure is not limited to the structure as shown in FIG. 7, and it is merely an example for illustrating one of the implements of the present disclosure.

(43) FIG. 8 depicts a schematic diagram of experimental data of an inductor device according to one embodiment of the present disclosure. As shown in the figure, the experimental curves of the quality factors of the inductor device not adopting the structural configuration of the present disclosure are C1, C2, and the experimental curves of the quality factors of the inductor device adopting the structural configuration of the present disclosure are C3, C4. As shown in the figure,

the inductor device adopting the structural configuration of the present disclosure has better the quality factors.

(44) Although the present invention has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein.

(45) It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present invention without departing from the scope or spirit of the invention. In view of the foregoing, it is intended that the present invention cover modifications and variations of this invention provided they fall within the scope of the following claims and their equivalents.

Claims

1. An inductor device, comprising: a first inductor, comprising: a first trace, wherein the first trace comprises a first spiral trace, wherein the first spiral trace is disposed on a first layer and a second layer; a second trace, wherein the second trace comprises a second spiral trace, wherein the second spiral trace is disposed on the first layer and the second layer; and a second connection member, disposed on a third layer, wherein the second spiral trace which is located on the second layer is coupled to the second spiral trace which is located on the first layer through the second connection member; a second inductor, comprising: a third trace; and a fourth trace, wherein the third trace and the fourth trace are symmetrical to each other; and a connection portion, configured to couple to the first inductor and the second inductor, wherein the connection portion comprises a first sub-connection portion, configured to couple to the second spiral trace which is located on the first layer and the first spiral trace which is located on the first layer.
2. The inductor device of claim 1, wherein the first inductor is located on the first layer and the second layer, and the second inductor is located on the second layer, and the connection portion is located on the first layer and the third layer.
3. The inductor device of claim 1, wherein the first inductor further comprises: a first input/output terminal, disposed on the first layer, and coupled to the first spiral trace which is located on the second layer, wherein the first spiral trace which is located on the second layer is coupled to the first spiral trace which is located on the first layer.
4. The inductor device of claim 3, wherein the first inductor further comprises: a first connection member, disposed on the third layer, wherein the first input/output terminal is coupled to the first spiral trace which is located on the second layer through the first connection member.
5. The inductor device of claim 1, wherein the first inductor further comprises: a second input/output terminal, disposed on the first layer, and coupled to the second spiral trace which is located on the first layer, wherein the second spiral trace which is located on the first layer is coupled to the second spiral trace which is located on the second layer.
6. The inductor device of claim 1, wherein the second spiral trace which is located on the first layer and the fourth trace which is located on the second layer are overlapped to each other partially.
7. The inductor device of claim 1, wherein the first spiral trace which is located on the first layer and the third trace which is located on the second layer are overlapped to each other partially.
8. The inductor device of claim 1, wherein the second inductor further comprises: a third input/output terminal, disposed on the third layer, and coupled to the fourth trace which is located on the second layer.
9. The inductor device of claim 8, wherein the second inductor further comprises: a fourth input/output terminal, disposed on the third layer, and coupled to the fourth trace which is located on the second layer.
10. The inductor device of claim 9, wherein in a central area of the inductor device, the third trace which is located on the second layer is coupled to the fourth trace which is located on the second

layer.

11. The inductor device of claim 10, wherein the second inductor further comprises: a center-tapped terminal, disposed on the third layer, and coupled to the third trace which is located on the second layer.

12. The inductor device of claim 11, wherein the center-tapped terminal is disposed symmetrically to the third input/output terminal and the fourth input/output terminal on a basis of the central area.

13. The inductor device of claim 10, wherein the connection portion further comprises: a second sub-connection portion, disposed on the first layer, wherein in the central area of the inductor device, the third trace, which is located on the second layer, is coupled through the second sub-connection portion to the fourth trace, which is located on the second layer.

14. The inductor device of claim 1, wherein the second inductor further comprises: a third connection member, disposed on the third layer, wherein the fourth trace, which is located on the second layer, is coupled through the third connection member to the fourth trace, which is located on the second layer, in an interlaced manner.

15. The inductor device of claim 14, wherein the second inductor further comprises: a fourth connection member, disposed on the third layer, wherein the fourth trace, which is located on the second layer, is coupled through the fourth connection member to the fourth trace, which is located on the second layer, in an interlaced manner, wherein the third connection member and the fourth connection member are located at different sides of the second inductor.

16. The inductor device of claim 1, wherein the second inductor further comprises: a fifth connection member, disposed on the third layer, wherein the third trace, which is located on the second layer, is coupled through the fifth connection member to the third trace, which is located on the second layer, in an interlaced manner.

17. The inductor device of claim 16, wherein the second inductor further comprises: a sixth connection member, disposed on the third layer, wherein the third trace, which is located on the second layer, is coupled through the sixth connection member to the third trace, which is located on the second layer, in an interlaced manner, wherein the fifth connection member and the sixth connection member are located at different sides of the second inductor.
